

Advanced (Habanero)

- Q1.** Given $f(x) = x^2$ and $g(x) = 2x - 1$, evaluate $f(g(2))$.
(A) 3
(B) 5
(C) 15
(D) 9
(E) 11
- Q2.** What is the domain of $f(g(x))$ when $f(x) = \frac{1}{x}$ and $g(x) = x + 1$?
(A) All real numbers
(B) All real numbers except -1
(C) All real numbers except 0
(D) All real numbers except 1
(E) None of the above
- Q3.** If $f(x) = x^2 - 1$ and $g(x) = x - 2$, what is $f(g(3))$?
(A) 5
(B) 7
(C) 8
(D) 10
(E) 12
- Q4.** Given $f(x) = \sqrt{x}$ and $g(x) = x^2$, what is the domain of $f(g(x))$?
(A) All real numbers
(B) All real numbers except 0
(C) Only non-negative real numbers
(D) Only positive real numbers
(E) None of the above
- Q5.** What is $f(g(-2))$ if $f(x) = x^2 + 1$ and $g(x) = 2x + 1$?
(A) 5
(B) 7
(C) 9
(D) 11
(E) 13
- Q6.** If $f(x) = \frac{1}{x^2}$ and $g(x) = x - 3$, what is the domain of $f(g(x))$?
(A) All real numbers
(B) All real numbers except 3
(C) All real numbers except 0
(D) All real numbers except -3
(E) None of the above
- Q7.** Given $f(x) = x^3$ and $g(x) = \frac{1}{x}$, evaluate $f(g(2))$.
(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) 1
(E) 2
- Q8.** If $f(x) = \sqrt{x+2}$ and $g(x) = x - 1$, what is the domain of $f(g(x))$?
(A) All real numbers
(B) All real numbers except -1
(C) All real numbers except -2
(D) Only real numbers greater than or equal to -1
(E) Only real numbers greater than or equal to -3

Open Ended Questions (FRQ)

- Q9.** A water tank can be filled at a rate of $f(t) = 2t$ gallons per hour for t hours. However, due to a leak, water also leaves the tank at a rate of $g(t) = t^2/10$ gallons per hour. Find and interpret $f(g(4))$. Provide your work and explanation.
- Q10.** For the functions $f(x) = x^2 - 4$ and $g(x) = x + 2$, determine the domain of $f(g(x))$. Explain your reasoning and provide any necessary work.

Chef's Tips

- Tip 1: Make sure students have access to a calculator and necessary formulas.
- Tip 2: Emphasize the importance of showing all work for free-response questions.
- Tip 3: Remind students to read each question carefully and manage their time effectively.